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with LTE connectivity. Sized at 10mm × 14mm, this module supports connectivity through secure digital input output (SDIO) or universal serial bus (USB), and can operate concurrently in client and micro-access point modes.

Bluetooth Smart with handy NFC. NINA-B1 BLE (Bluetooth Smart), a standalone module with integrated NFC capability, was recently announced by U-Blox. Comprising an antenna, radio transceiver, an embedded ARM Cortex M4F MCU and a BLE stack, the chip is design-ready for applications. The application memory of 512kB flash and 64kB RAM allows for future firmware upgrades to be performed over the air.

Other strong competitors like Texas Instruments CC2540, 2541 and 2564 have been popular in the market for quite some time now. Featuring an 8051 processor, 256kB flash and 8kB RAM, these handy chipsets support a 12-bit ADC, comparator, temperature sensor and battery monitor, hence reducing the cost on several peripherals.

Data transfer speed. Manufacturers have been trying to increase data transfer speeds for a while now. Entities like Intel, Qualcomm and Hitachi are working on yet another faster Wi-Fi standard, 802.11ad. With an ideal speed of about 7Gbps, transmission speeds are expected to reach a new high.

In recent news, Intel and Qualcomm claim to have gotten their respective 802.11ad (WiGig) chips and antennae to successfully link up. This offers speeds of up to 4.6Gbps under ideal conditions. Intel WiGig is very handy in removing the hassle of wires all over the desk. A docking station supports an Ethernet port and connectivity for an external Internet connection.

Hi-speed NFC. NTAG I2C plus NFC tags from NXP Semiconductors combine a passive NFC interface with a contact I2C interface. Features in the chip include hi-speed transfer, password protection, full memory-access configuration from both interfaces, an originality signature for protection against cloning and four times higher pass-through performance, along with energy-harvesting capabilities.

PN7462 family of single-chip NFC reader solutions from NXP combine a microprocessor with USB, for contact and contactless interfaces. According to a press release by NXP, "It is specifically designed for use in readers in applications including corporate access, hospitality or home banking."

Employing the technology

Selection of a particular technology requires consideration of several parameters. According to Karnataki, "Power consumption, communication range, addressing

methods and user-friendly configuration software are some important features to keep in mind while selecting a communication chip for designing a system."

From a designer's point of view, "TTL compatible I/O interface avoids the involvement of intermediate data converters," he adds.

An important factor in employing any technology is the operating environment. The chipset selected must be capable of delivering data despite noise and other constraints. Employing the technology also depends on use cases: How the use case will be catered to, with a minimal number of components.

Anand adds, "For wearables, you are not going to design it on a hard PCB; you need a flexible one." Unny supports the statement with, "Wearables have ensured large-scale adoption of wireless technologies. This has significantly brought down the cost of these chipsets and made these technologies available to the masses."

Looking ahead

Unny states, "Wireless technologies were limited to industrial environments in the past. Wearables and smartphones have made these technologies available to the general public. Secure payments, object tracking, health and fitness tracking, and human security are some of the most significant use cases."

Newer standards are being developed for Wi-Fi, which offer higher speeds than ever before. Bluetooth research has been focused on reducing energy requirements. Monetary transactions have been aided in part by NFC. After all, self-checkout lanes at Wal-Mart are the best ones, and if you could pay with just a tap of your mobile instead of swiping a card and entering a pin, the lanes would certainly move faster. **EFY**